

Description

Physical examination, CT scan of the head and neck (ICD-10 Procedure Code: BW290ZZ Computerized Tomography (CT Scan) of Head and Neck using High Osmolar Contrast) reveal a fracture to the base of the patient's skull and massive intracranial bleeding. While the patient's examination and diagnostic tests are proceeding an emergency room tech uncovers the patient's driver's license among the belongings brought to the ED by the paramedics along with a flight stub showing she traveled from Paris 2 days earlier. At 5:20 PM the patient's medical record is updated with working diagnosis ICD-10 codes S02.112A (Type III occipital condyle fracture, initial encounter for closed fracture) and S06.0X9A (Concussion with loss of consciousness of unspecified duration, initial encounter). Also at this time, the patient's license is used to update the medical record with name and age.

NIST Hospital's Emergency Department reports syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) at least once per day. At 5:30 PM on July 17, 2024, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT 08 Update message about this encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity.

Comments

The dates and times in this test case illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the test case with an HIT Module, only date and time format will be validated within tester submitted test data.

Pre-condition

A04-Registration message is sent before the A08-Update message.

Post-Condition

No Post-Condition

Test Objectives

This test step examines an HIT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Emergency Department.
- Chief complaint is captured as narrative in PV2-3 Admit Reason.
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Adult case where age is measured in years.
- Updates to patient demographics
- Addition of Diagnoses and Procedure
- Insurance is known

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
S02112A	Type III occipital condyle fracture, initial encounter for closed fracture	I10C	S02112A^Type III occipital condyle fracture, initial encounter for closed fracture^I10C	Yes
721321003	Fracture of occipital condyle of skull type III (disorder)	SCT	S02112A^Fracture of occipital condyle of skull type III (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
S02.112A	Type III occipital condyle fracture, initial encounter for closed fracture	I10C	S02.112A^Type III occipital condyle fracture, initial encounter for closed fracture^I10C	Yes
S02112A	Type III occipital condyle fracture, initial encounter for closed fracture	I10	S02112A^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No
S02.112A	Type III occipital condyle fracture, initial encounter for closed fracture	I10	S02.112A^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No
127281000	Fracture of occipital condyle (disorder)	SCT	127281000^Fracture of occipital condyle (disorder)^SCT	Yes

DG1(2)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
S060X9A	Concussion with loss of consciousness of unspecified duration, initial encounter	I10C	S02112A^Concussion with loss of consciousness of unspecified duration, initial encounter^I10C	Yes
110030002	Concussion injury of brain (disorder)	SCT	110030002^Concussion injury of brain (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
S06.0X9A	Concussion with loss of consciousness of unspecified duration, initial encounter	I10C	S02.112A^Type III occipital condyle fracture, initial encounter for closed fracture^I10C	Yes
S060X9A	Concussion with loss of consciousness of unspecified duration, initial encounter	I10	S02112A^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No
S06.0X9A	Concussion with loss of consciousness of unspecified duration, initial encounter	I10	S02.112A^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No

62564004	Concussion with loss of consciousness (disorder)	SCT	62564004^Concussion with loss of consciousness (disorder)^SCT	Yes
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The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

Notes

ICD-10 CM codes are provided in the Test Data and required for coding the diagnoses in this Test Case. If the HIT developer uses different but equivalent ICD-10 CM codes than the ones provided, the Tester may ignore errors generated by the Test Tool related to incorrect code when the ICD-10 CM code data used in the message are determined via visual inspection to be valid data.

ICD-10 CM codes are acceptable with or without decimals.

Messages in this test case will be used in a Manual Test Step under functional requirement testing.